

MODEL QUESTION PAPER

REVISION 2010
SUB CODE:3059

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

DIGITAL ELECTRONICS
(Common to AE, BM, EA, EL, EC, EP, MD, TC)

Time: 3 Hrs

Maximum Marks :100

PART A

{Answer all questions. Each question carries 2 marks}

I

1. What is BCD code? Write BCD code for 25.
2. Define propagation delay time for a digital logic family.
3. What is a sequential logic circuit?
4. What is meant by 'Race around condition' in JK Flip-flop?
5. Define Resolution in Digital to Analog Converter. (5x2=10)

PART B

(Answer any 5 questions. Each question carries 6 marks)

II

1. Convert the given decimal number to binary and hexadecimal.
367.25
2. Discuss ASCII and EBCDIC codes.
3. Explain the working of TTL inverter with diagram.
4. With Circuit diagram and Truth table explain the operation of a 4to1 Multiplexer.
5. Explain the working of a JK Flip Flop using NAND gates. Give Truth Table.
6. Give the classification of Shift Registers according to inputs given and outputs taken.
7. Explain NVRAM. (5x6=30)

PART C

(Answer one full question from each module. Each question carries 15 marks)

MODULE 1

IIIa Explain Hamming code with Example. (5)

b. Discuss Gray codes and Excess 3 codes. (5)

c. List the advantages and disadvantages of Karnaugh Map (5)

OR

IV a Using Karnaugh map reduce the given expression (9)

$$f = \sum m(0, 2, 5, 7, 8, 10, 13, 15).$$

b. Simplify the given expression using Boolean algebra (6)

$$\overline{A}B + A + BC + \overline{AB} + \overline{BC}$$

MODULE II

V.a Define the terms used in digital logic families.

Noise Margin, Fan in, Fan out, Speed Power product, Sourcing and Sinking currents (15)

OR

VI a Explain with circuit diagram the working of Decimal to BCD Encoder (8)

b Explain the working of Half adder and Half subtractor (7)

MODULE III

VII a. Describe the working of a Master Slave JK Flip Flop with circuit diagram and Truth table (10)

b. Distinguish between Asynchronous and Synchronous sequential logic circuits. (5)

OR

VIII a. Design and implement MOD 8 synchronous counter using JK Flip Flop (15)

MODULE IV

- IX a** Differentiate SD RAM and EDO RAM **(8)**
- b Explain the working of counter type Analog to Digital converter **(7)**

OR

- X a** State the principle of working of ROM **(5)**
- b. Explain the working of R-2R ladder network type Digital to Analog Converter **(10)**